



# Introduction to the Program Design and Programming

ICT 1305

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CHAPTER 01

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**GETTING**

**TO**

**KNOW**

**YOU**

# Introduction to the Course

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- Course Capsule
- Assignment Criteria
- Practical Classes
- Course Design
- How to use Google classroom

# Course Content

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- **Techniques of Problem Solving:** Algorithms, Flowcharts, Pseudo Codes, Paper Simulations.
- **Fundamentals of Programming:** Structure of a Program, Input/Output, Variable Declaration, Arithmetic Operations, Relational Operations, Logical Operations.
- **Control Structures:** Conditional statements, Repetition statements , Break and Continue statements, Sub programs
- **Data handling:** Scope of Variables and Parameters, Recursion, Arrays, Records, Comparison of structure and class, Dynamic memory allocation, Best programming practices.



# Learning Outcomes

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At the end of the course, the student should be able to,

- Demonstrate the knowledge of programming fundamentals
- Demonstrate a basic knowledge in programming best practices
- Design, write and test programs according to a basic requirement specification



# Evaluation Criteria

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Continuous assessment - 40%

Mini project - 20%

Assignments - 20%

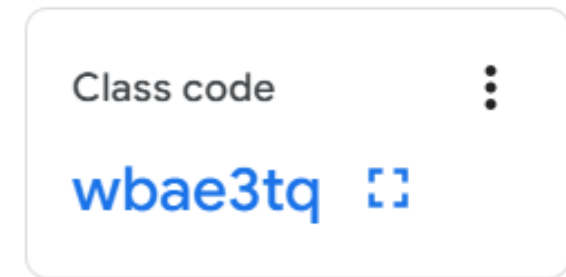
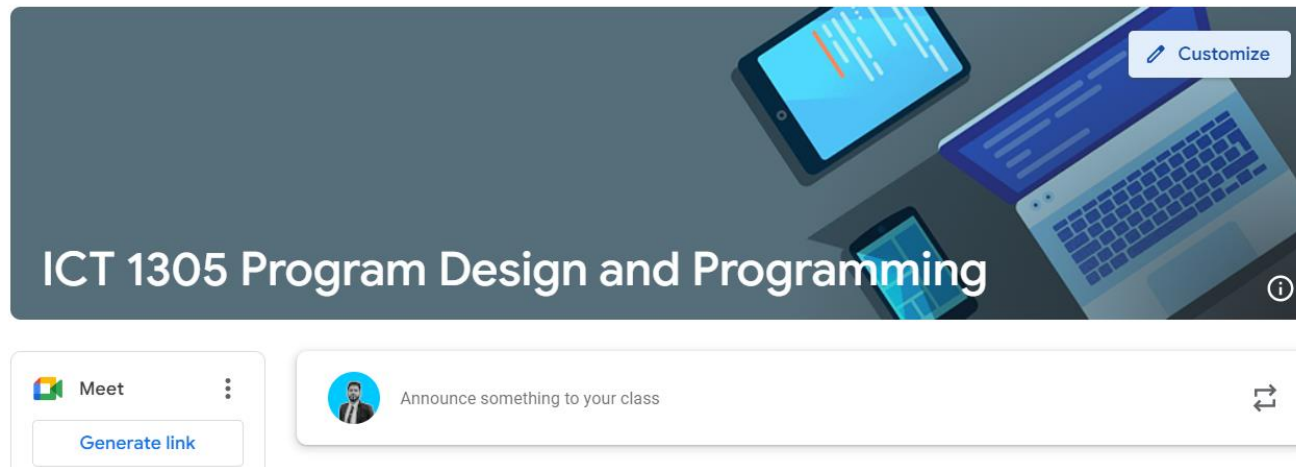
End semester examination - 60%

Practical exam - 20%

Written exam - 40%

# Google Classroom

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Link : <https://classroom.google.com/c/NzI5NTU3NDc4MDkx?cjc=wbae3tq>

Any Questions ???

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# Problem Solving

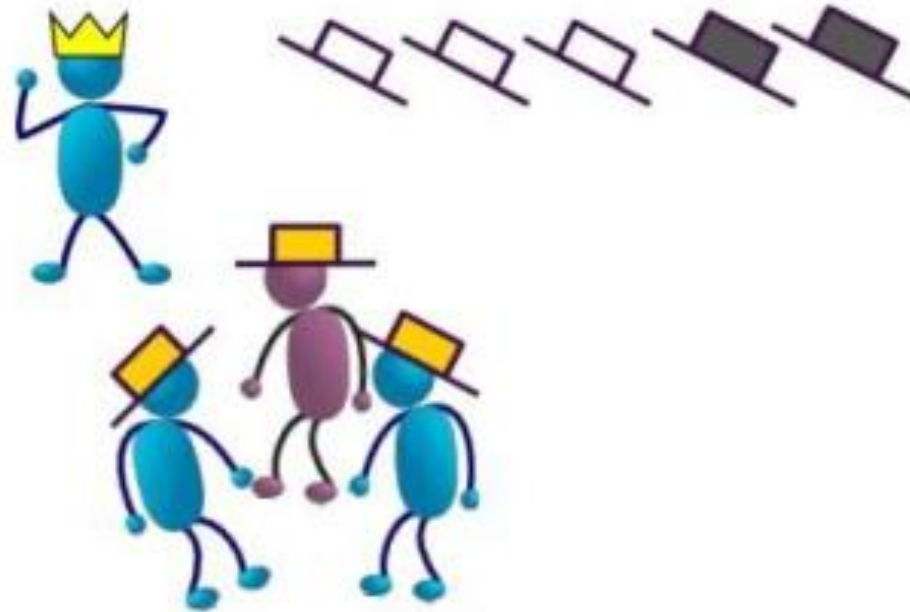
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- Problem solving, or breaking down the solution to a problem into sequential steps is by far the most difficult part of computer programming
- There are many activities we can engage in that will help sharpen our problem solving skills.



# The Three Hats

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# The Three Hats

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- Three men are condemned to die, one of whom is blind.
- The king decides he will offer them an opportunity to be set free.
- The three men are arranged in a circle facing one another.
- The king produces 5 hats: 2 black and 3 white.
- The king places a hat on the head of each person and then destroys the two remaining hats.
- The men have no idea which hats have been destroyed.
- The king instructs them, “The first one of you who can tell me the color of his hat will be set free.”

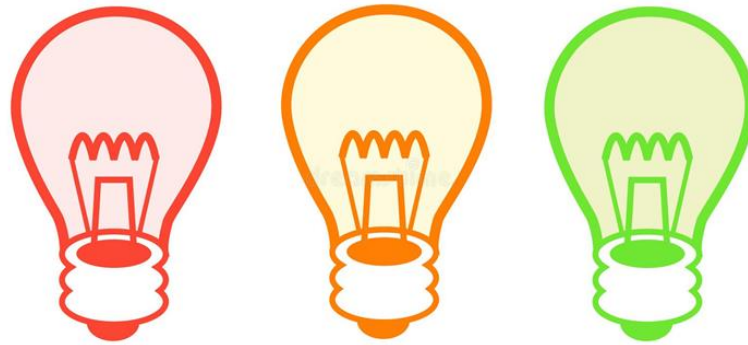
# The Three Hats

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- A period of time passes in silence and then finally the blind man tells the king the color of his hat and is set free.
- What color hat was the blind man wearing and how did he know?
- This does not have a “trick” answer – your answer should be very logical and well thought out.
- Be able to explain your answer from the viewpoint of each of the three prisoners.
- Hint: Each sighted man can see the blind man’s hat as well as that of the other sighted man.
- What does the pause in time infer?

# Three Bulbs

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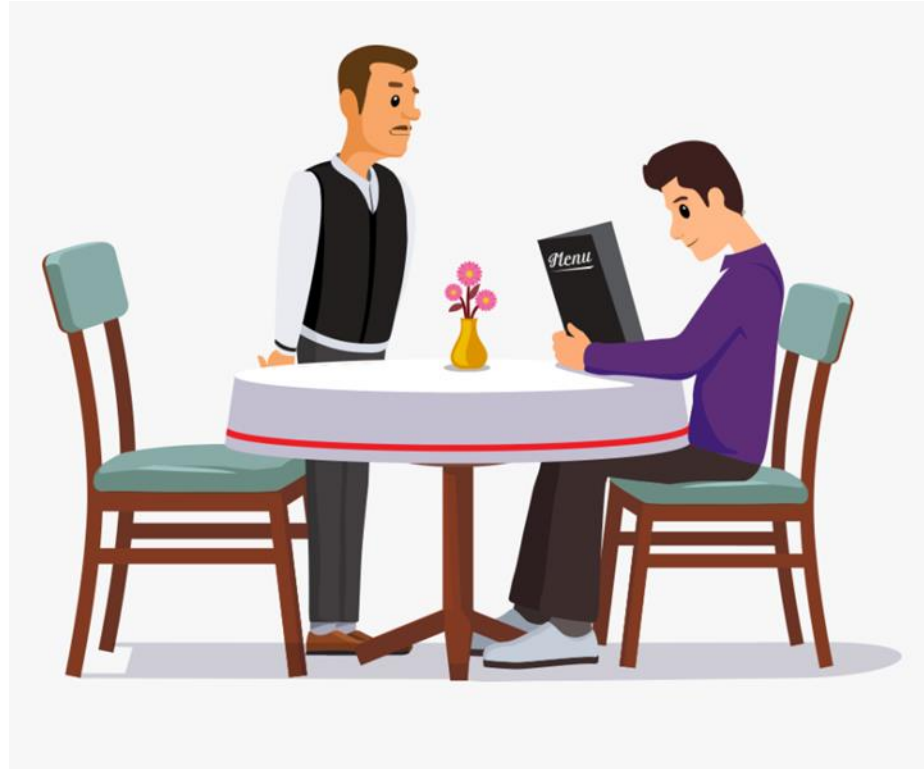
# Three Bulbs

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- You are standing next to three switches. You know these switches belong to three bulbs in a room behind a closed door – the door is tight closed, and heavy which means that it's absolutely impossible to see if any bulb is on or not.
- All three switches are now in position off.
- You can do whatever you want with the switches and when you are finished you open the door and go into the room.
- While in there you have to tell which switch belongs to which bulb.
- How will you do that?

# The Waiter

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# The Waiter

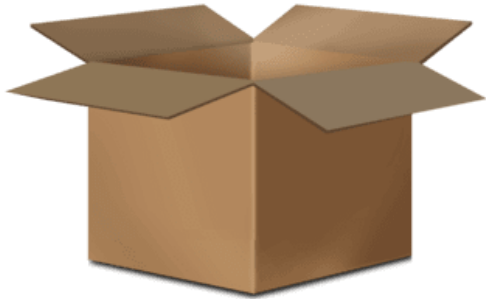
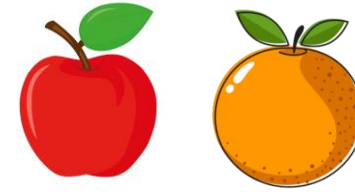
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- Three men in a cafe order a meal the total cost of which is \$15. They each contribute \$5.
- The waiter takes the money to the chef who recognizes the three as friends and asks the waiter to return \$5 to the men.
- The waiter is not only poor at mathematics but dishonest and instead of going to the trouble of splitting the \$5 between the three he simply gives them \$1 each and pockets the remaining \$2 for himself.
- Now, each of the men effectively paid \$4, the total paid is therefore \$12. Add the \$2 in the waiters pocket and this comes to \$14.....where has the other \$1 gone from the original \$15?



# The Box

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# The Box

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- There are three boxes. One is labeled "APPLES" another is labeled "ORANGES". The last one is labeled "APPLES AND ORANGES". You know that each is labeled incorrectly.
- You are asked to pick one fruit from one box which will you choose?
- How can you label the boxes correctly?

# Miscellaneous Problems

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- A mechanic's brother won the car race. But the man who won had no brother.  
How is this possible?

# Miscellaneous Problems

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- A farmer has some chickens and some goats. Together there are 43 heads and 108 legs. How many chickens does the farmer have? How many goats?

# Thank You

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